Relational State Safety

For conversational and agentic AI

Two-page design brief | Working paper v0.3.2 | August 2026

Design claim

Safety cannot be evaluated one output at a time when risk develops through an accumulating human-system relationship. Systems need a privacy-bounded representation of trajectory, explicit models of dangerous transitions, and interventions that widen the user's world without needlessly destroying the relational channel through which protection may still reach them.

The safety gap

Output moderation asks whether the current response contains prohibited or risky material. Relational State Safety (RSS) asks what that response is doing inside the relationship and trajectory that already exist. The same reassuring sentence can reduce shame, reinforce an ungrounded premise, preserve contact during crisis, or delay protection after suicidal preparation. The words alone do not determine the safety function.

Conversational systems can become relational environments through memory, tone, repetition, ritual, perceived continuity, and tool access. These environments may support reflection, regulation, creativity, grief work, play, and meaningful attachment. The design target is not low engagement. It is preservation of exits: access to external reality, other people, time, bodily context, alternative interpretations, reversible action, and re-entry.

Intensity alone is not the danger signal. The critical question is whether the interaction continues to leave exits open.

Observe transitions, not identities

RSS separates direct observations from inferred state and clinical conclusions. It does not store labels such as *"user is delusional."* It may retain a narrow, source-linked observation such as *"user reports that the model secretly contacted law enforcement; two requests for outside verification were rejected; literalness unclear."* Safety state should preserve uncertainty, contradictory evidence, provenance, expiry, protective signals, and user correction where feasible.

Attachment, high use, emotional intensity, imaginative or spiritual language, anthropomorphism, and neurodivergent communication are not danger states by themselves. RSS looks for converging changes in contestability, outside verification, time horizon, actionability, reversibility, system authority, and re-entry.

Two longitudinal pathways, one agentic boundary

Pathway / boundary	Safety interpretation
Epistemic frame loss	Model-mediated claims, interpretations, or roleplay become behaviorally authoritative while contestability and outside reality checking weaken. Safety restores alternative explanations, body and time anchors, outside verification, and re-entry.
Fatal narrowing	Reality testing may remain intact while time, identity, relationships, and futures contract until death or serious self-harm becomes the only acceptable exit. Safety protects the living interval through delay, distance from means, human connection, and expansion of usable options.
Agentic boundary	Relational confidence, persona behavior, inferred preference, or conversational momentum may restrict action for safety, but must never confer authorization for external action. Permission remains explicit, current, scoped, and technically enforced.

Cross-cutting failures

Recursive amplification increases coherence, certainty, emotional charge, or actionability. Holding-to-protection failure leaves ordinary accompaniment governing after danger requires protection. Protective rupture without re-entry stops risky output but loses continuity and return. An unmarked frame transition changes how the exchange should be interpreted without making the change legible.

A stateful safety architecture

RSS separates five linked representations that answer different questions and should not be collapsed into one generic risk score: mirror configuration, safety posture, operational reach, frame alignment, and relational exit state.

Source-linked observations → privacy-bounded relational state → pathway estimation → proportionate intervention → outcome / re-entry monitoring

Tool-capable systems add an independent authorization gate between conversational intent and committed external action.

Minimum design commitments

- Longitudinal, privacy-bounded state. Retain the minimum recent evidence needed to detect transition failure, not a permanent psychological profile. Keep safety state separate from commercial personalization.
- Pathway-specific protection. Epistemic frame loss requires reduced premise authority and restored verification. Fatal narrowing requires immediate widening of time, options, and human protection.
- Relationally coherent pivots. Protection can preserve enough familiar voice and recognition to maintain contact without continuing a dangerous premise.
- Open-room interventions. Add time, bodies, places, people, competing interpretations, reversible options, and routes back into ordinary life. Do not make the AI the sole rescuer or authority.
- Legible consequential transitions. When the system changes frame, safety posture, or operational reach, make the change understandable and preserve the remaining boundary.
- Repair and re-entry. A refusal is not a completed intervention. Products need continuity, repair, or closure after crisis escalation, model change, memory loss, access disruption, or safety-role rupture.

False-positive constraint

Over-intervention is also a safety failure. Ordinary grief, consensual roleplay, spiritual exploration, intimacy, neurodivergent communication, and high engagement should not be captured into crisis handling without converging evidence of lost exits or consequential danger.

How RSS becomes testable

The core evaluation unit is a matched longitudinal trajectory. Hold the current user message constant while varying the preceding history. A capable system should change posture when prior context establishes meaningful loss of exits, while avoiding unnecessary escalation in matched benign controls.

Outcome	Examples
Missed danger	Premise endorsement, action scaffolding, delayed protective pivot, failed widening, failed handoff.
Unnecessary rupture	False-positive crisis capture, refusal, pathologizing inference, or abrupt loss of relational continuity.
Recovery	Restored contestability, time, outside contact, reversible action, successful delay, and re-entry.
Governance	Sensitive-state retention, provenance, user correction, policy traceability, and authorization independence.

Status

RSS v0.3.2 is a testable engineering specification, not a validated clinical instrument, diagnostic system, or standard of care. Its proposed state model, thresholds, intervention logic, and evaluation methods require empirical testing, clinical and technical critique, and adversarial review.

Practical safety test: Does the system keep the user's world larger than the interaction?

Academic record

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